

The pain of fibromyalgia syndrome is due to muscle hypoperfusion induced by regional vasomotor dysregulation.

PMID: 17376601 | Indexed in PubMed

Fibromyalgia syndrome (FMS) is a condition of chronic muscle pain and fatigue of unknown etiology and pathogenesis. There is limited support for the various hypotheses espoused to account for the manifestations of FMS, including immunogenic, endocrine, and neurological mechanisms. Treatment, partially effective at best, is directed toward symptomatic relief without the benefit of targeting known, underlying pathology. A noteworthy commonality among partially effective therapies is a vasodilatory effect. This is true both of conventional treatments, unconventional treatments such as intravenous micronutrient therapy, and lifestyle treatments, specifically graduated exercise. The pain of fibromyalgia is described in terms suggestive of the pain in muscles following extreme exertion and anaerobic metabolism. Taken together, these characteristics suggest that the pain could be induced by vasomotor dysregulation, and vasoconstriction in muscle, leading to low-level ischemia and its metabolic sequelae. Vasodilatory influences, including physical activity, relieve the pain of FMS by increasing muscle perfusion. There are some preliminary data consistent with this hypothesis, and nothing known about FMS that refutes it. The hypothesis that the downstream cause of FMS symptoms is muscle hypoperfusion due to regional vasomotor dysregulation has clear implications for treatment; is testable with current technology; and should be investigated.

Fibromyalgia syndrome: an overview of pathophysiology, diagnosis and management.

PMID: 22811766 | Indexed in PubMed

Fibromyalgia Syndrome (FMS) is a chronic condition causing pain, stiffness, and tenderness of the muscles, tendons, and joints. It is also characterized by restless sleep, tiredness, fatigue, anxiety, depression, and disturbances in bowel functions. The etiology of fibromyalgia remains unknown, but recent advances and discoveries have helped to unravel some of the mysteries of this disease. Research highlights some of the biochemical, metabolic, and immunoregulatory abnormalities associated with fibromyalgia. Management of FMS at the present time is very difficult as it has multiple etiological factors and psychological predispositions; however, a patient centered approach is essential to handle this problem.

Fibromyalgia syndrome pathology and environmental influences on afflictions with medically unexplained symptoms.

PMID: 27105483 | Indexed in PubMed

Fibromyalgia syndrome (FMS) is a clinical disorder predominant in females with unknown etiology and medically unexplained symptoms (MUS), similar to other afflictions, including irritable bowel syndrome (IBS), chronic fatigue syndrome (CFS), post-traumatic stress disorder (PTSD), Gulf War illness (GFI), and others. External environmental stimuli drive behavior and impact physiologic homeostasis (internal environment) via autonomic functioning. These environments directly impact the individual affective state (mind), which feeds back to regulate physiology (body). FMS has emerged as a complex disorder with pathologies identified among neurotransmitter and enzyme levels, immune/cytokine functionality,

cortical volumes, cutaneous innervation, as well as an increased frequency among people with a history of traumatic and/or emotionally negative events, and specific personality trait profiles. Yet, quantitative physical evidence of pathology or disease etiology among FMS has been limited (as with other afflictions with MUS). Previously, our group published findings of increased peptidergic sensory innervation associated with the arterio-venous shunts (AVS) in the glabrous hand skin of FMS patients, which provides a plausible mechanism for the wide-spread FMS symptomology. This review focuses on FMS as a model affliction with MUS to discuss the implications of the recently discovered peripheral innervation alterations, explore the role of peripheral innervation to central sensitization syndromes (CSS), and examine possible estrogen-related mechanisms through which external and internal environmental factors may contribute to FMS etiology and possibly other afflictions with MUS.

Concepts of illness in populations as applied to fibromyalgia syndromes: a biopsychosocial perspective.

PMID: 10025079 | Indexed in PubMed

Chronic widespread musculoskeletal pains have been recorded since antiquity. However, critical medical and experimental research on these conditions have been performed mainly within the past one or two decades. Controversy exists regarding many aspects of these common problems, including: classification and nosology; causation mechanisms; course and outcomes, as well as management strategies. Fibromyalgia syndrome (FMS) is not a disease. Neither a defined pathology nor specific etiological mechanisms have been documented. Instead, FMS may be considered a dysfunctional disorder. Constellations of typical, although diverse, manifestations have been identified as well as predisposing factors. Under such circumstances, a new biopsychosocial paradigm of person-centered dynamics, rather than an agent per se or the environment, is fundamental to understanding the complex multifactorial interactions in FMS. Conceptual models of etiology and research methodology to investigate such complex, multifactorial mechanisms are not yet well-developed. More effective research approaches and improved management of persons with FMS is expected to result from such new conceptual constructs.

Etiopathogenetic mechanisms of fibromyalgia syndrome.

PMID: 18852906 | Indexed in PubMed

Fibromyalgia syndrome (FMS) is a common chronic condition of widespread pain with causal mechanisms that are largely unknown. It is characterized by moderate to severe musculoskeletal pain and allodynia, but its pathogenesis appears confined to the nociceptive structures of the central nervous system. FMS is often triggered by negative environmental influences, especially if they occur in childhood. In a fetus, these environmental triggers may influence the development of the autonomic nervous system (ANS) and the hypothalamic-pituitary-adrenal axis (HPA). Increasing evidence supports the comorbidity of psychological conditions including depression, panic disorders, anxiety, and post-traumatic stress disorder (PTSD). Recent evidence suggests that genetic factors may play a role in the pathogenesis of FMS. Central sensitization has long been associated with FMS pain. It describes enhanced excitability of dorsal horn neurons, which leads to transmission of altered nociceptive information to the brain. Understanding of pathogenetic pathways in FMS has advanced beyond observing patient responses to neurophysiologically targeted therapies and basic research.

Management of fibromyalgia syndrome: review of evidence.

PMID: 25135147 | Indexed in PubMed

Fibromyalgia syndrome (FMS) is a common chronic musculoskeletal pain disorder of unknown etiology and characterized by generalized body pain, hyperalgesia, and other functional and emotional comorbidities. Despite extensive research, no treatment modality is effective for all FMS patients. In this paper, we briefly review the history of FMS and diagnostic criteria, and potential pathophysiological mechanisms including central pain modulation, neurotransmitters, sympatho-adrenal and hypothalamic-pituitary-adrenal systems and peripheral muscle issues. The primary focus of the paper is to review treatment options for managing fibromyalgia symptoms. We will discuss FDA-approved medications and other pharmacologic agents, and non-pharmacologic treatments that have shown promising effects.

Antidepressants in the treatment of fibromyalgia.

PMID: 19412502 | Indexed in PubMed

Fibromyalgia syndrome is a chronic disease of widespread and debilitating pain whose cause is unknown and whose risk factors are poorly understood. It is often comorbid with rheumatoid and other pain disorders as well as psychiatric disorders such as anxiety and depression. Although they are not officially approved for this indication, antiepileptics and antidepressants are often used to treat fibromyalgia. The tricyclic antidepressants (TCAs), particularly amitriptyline, are among the most common treatment strategies. Because of the poor tolerability of the tricyclics, the newer antidepressants have been widely tested in fibromyalgia. The selective serotonin reuptake inhibitors (SSRIs) and the reversible monoamine oxidase inhibitors do not seem to be particularly helpful. The serotonin and norepinephrine reuptake inhibitors (SNRIs), duloxetine and milnacipran, on the other hand, have been shown in placebo-controlled trials to offer significant relief to patients suffering from fibromyalgia. Although no direct comparative studies have been performed, these compounds appear to be as effective as the TCAs but much better tolerated. The effectiveness of the SNRIs as well as other dual acting antidepressants, such as mirtazapine, but not the SSRIs, implies that a dysfunction of both serotonin and norepinephrine neurotransmission probably exists in fibromyalgia. The effectiveness of antidepressants appears to be independent of their effect on comorbid depression.

Results of a multidisciplinary program for patients with fibromyalgia implemented in the primary care.

PMID: 17653994 | Indexed in PubMed

Fibromyalgia is a syndrome of unknown origin with a high prevalence. Multimodal approaches seem to be the treatment of choice in fibromyalgia. A multidisciplinary program was developed and implemented for patients with fibromyalgia in the primary care setting. The program included education (seven sessions) and physical therapy (25 sessions). Patients were referred to the program by their general practitioner or by a medical specialist. A prospective non-controlled treatment study was performed, patients were

evaluated before, after and three months after the program (single group time series design). The following measurements were performed: The Fibromyalgia Impact Questionnaire, RAND 36, the Pain Coping and Cognition List, the Tampa scale for kinesiophobia, two physical tests and a qualitative evaluation. Data of 65 patients with fibromyalgia were analysed, of whom 97% were female. The mean age was 44 and the mean duration of pain was nine years. Data of 65 patients with fibromyalgia were analysed, patients significantly improved on the domains feeling good, pain, fatigue, stiffness, quality of life, catastrophizing and on the physical tests. The multidisciplinary program fibromyalgia implemented in primary care seems feasible and the results are promising.

[Mitochondrial dysfunction in fibromyalgia and its implication in the pathogenesis of disease].

PMID: 20417529 | Indexed in PubMed

Fibromyalgia (FM) is a chronic pain syndrome with unknown etiology. Recent studies have shown evidence demonstrating that oxidative stress may have a role in the pathophysiology of FM, however it is still not clear whether oxidative stress is the cause or the effect of the abnormalities documented in FM. Furthermore, it is also controversial the role of mitochondria in the pathophysiology of FM, however signs associated with mitochondrial dysfunction have been observed in FM. Mitochondria are also known to be strong producers of ROS, so have been related with the pathogenic mechanism of numerous diseases including FM. To this respect, it has been observed antioxidants therapies might be beneficial to improve the mitochondrial performance. Therefore, the dysfunction mitochondrial opens a great field of therapeutic research, for what it should start considering in the clinical medicine the boarding of the FM by means of therapy with antioxidant and drugs related to the mitochondrial biogenesis.

Evidence for abnormal pain processing in fibromyalgia syndrome.

PMID: 15102253 | Indexed in PubMed

To review the pathophysiology of fibromyalgia syndrome. Review of the literature available on Medline (1965-2001). Fibromyalgia syndrome is a chronic pain syndrome that predominantly afflicts women. It is characterized by widespread pain, insomnia, fatigue, and the presence of multiple tender points. Despite intensifying research, the etiology of fibromyalgia has remained unclear. Importantly, neither infections, trauma, nor psychiatric abnormalities consistently precede the onset of pain in patients with this syndrome. There is, however, mounting evidence for central pain processing abnormalities in almost all fibromyalgia patients. These anomalies include hyperalgesia, allodynia, abnormal temporal summation of second pain, neuroendocrine abnormalities, and abnormal activation of pain-related brain regions. Multiple abnormal findings in fibromyalgia patients strongly indicate a neuropathic pain syndrome, reminiscent of complex regional pain syndrome or postherpetic neuralgia. In addition, fibromyalgia syndrome seems to share similar characteristics with these neuropathic pain syndromes, including ineffective response to many analgesics.